

4. Let X_1 be $N(\mu, 1)$. Let X_2 be $N(\mu, 2)$ and independent of X_1 . The problem is to estimate μ . Consider the following three estimators:

$$T_1 = X_2, \quad T_2 = \frac{X_1 + X_2}{2}, \quad T_3 = \frac{1}{5}X_1 + 1.$$

(a) Which estimators are unbiased?

(b) Find the mean square error of each estimator. If $\mu = 1$, which estimator is preferred?